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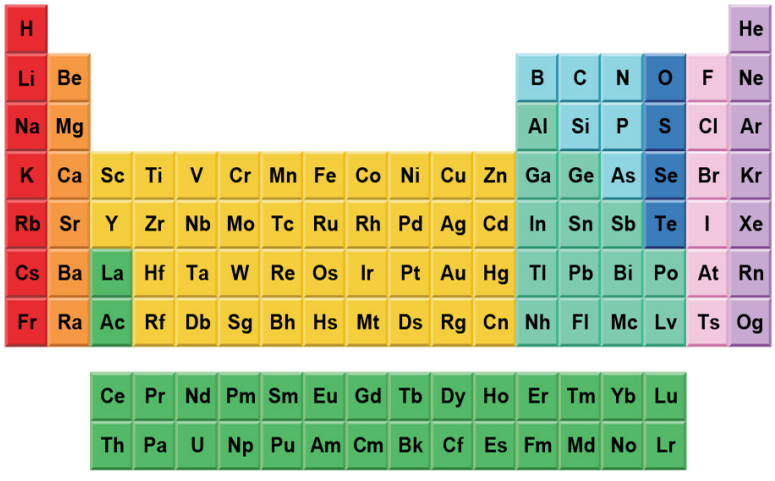
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Performance Enhancement of PV Module with Cooling Effect using Wet Wicks

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Abstract. Solar energy is one of the most potential sources of power production among all the renewable energies which is eco-friendly and can be much cheaper too. The energy conversion efficiency of the PV module decreases even more due to an increase in operating cell temperature over a certain limit. One way of improving the efficiency of the system is to maintain a low operating temperature by cooling it down during its operational period. This study compares the effects of cooling on the performance of photovoltaic systems. Experiments are performed on the solar panel inclined at a fixed 23.45˚ angle with the horizontal to the south without active cooling initially to have a set of reference performance parameters for comparison. The polycrystalline PV Module was tested with different cooling methods including Normal water Spray over the front surface, wet jute, and wet woolen material over the back surface of the panel. The cooling of the panels combined with the soiling effect enhances solar efficiency. The wet woolen material results in the most significant efficiency improvement, followed closely by wet jute and normal water spray.

Keywords: Solar Energy; PV cell; Water-cooling of PV cell; Solar Efficiency.

1. Introduction

Temperature is the main factor in increasing and decreasing efficiency (Dubey et al., 2013). PV modules convert only 4-17% of the increasing solar radiation in electricity. From that 50% is used to produce heat which increases the temperature. The PV panel loses efficiency if the temperature exceeds 25°C (Idoko et al., 2018). So maintaining 25°C is a must to have efficiency in a good range. After achieving comparatively a higher temperature, both the band gap of a semi-conductor and voltage decrease. The opposite phenomenon is with the decrement in the temperature where current and voltage increase. The temperature coefficient is that if a panel is operating in temperatures lower than 25°C, the temperature coefficient will be positive, and the solar panels will increase in efficiency. The ideal conditions for maximum solar energy production are sunny days with cooler temperatures.

In addition, some soil patches such as leaves, bird droppings, and dirt patches that block cells of a PV module, have a severe effect on PV modules. Water spray over the front surface of modules removes the dust and increases the electrical power output. There are different techniques to cool the PV module and enhance its efficiency including wet jute wicks attached directly to the back panel (Bhakre et al., 2023), wet cotton (Alktranee & Bencs, 2023), heat pipes (Mim et al., 2024, Hassan et al., n.d),

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different shape memory polymers for their high enthalpy storage (Zaman et al., 2023) and floating PV module (Rahaman et al., 2023).
In this study, we integrate and compare frontal water cooling, soiling effect (Schill et al., 2015), and wet wick methods on PV modules. We chose jute and wool based on their widespread availability and their natural wicking properties. Preliminary studies indicated these materials to be promising in terms of moisture regain and moisture content (Ali et al., 2015, 2017; Lv et al., 2014). These two materials represented a good balance of cost, availability, and performance compared to other methods available in the literature.

Table 1. Different Wick Structures

Fiber Name	Moisture Regain (MR %)	Moisture Content (MC %)
Cotton	8.5	7.34
Jute	13.75	12.1
Silk	11.0	9.91
Viscose	11.0	9.91
Wool	16.0	13.8

Firstly, we discussed the recorded electrical output values due to the cooling of the panels. Then the solar efficiency in different conditions. Finally, compared and analyzed the results with supporting theory and plots.

2. Method

2.1. Experimental Setup

We selected two Polycrystalline PV Modules of 50 watts with a temperature coefficient of -0.44/°C The values of short circuit current (Isc) and open circuit voltage (Voc) are 3.17Amp and 21.8 Volt respectively. The area of the panel is 0.45 m2, so Jute and woolen fabrics were fabricated according to this.

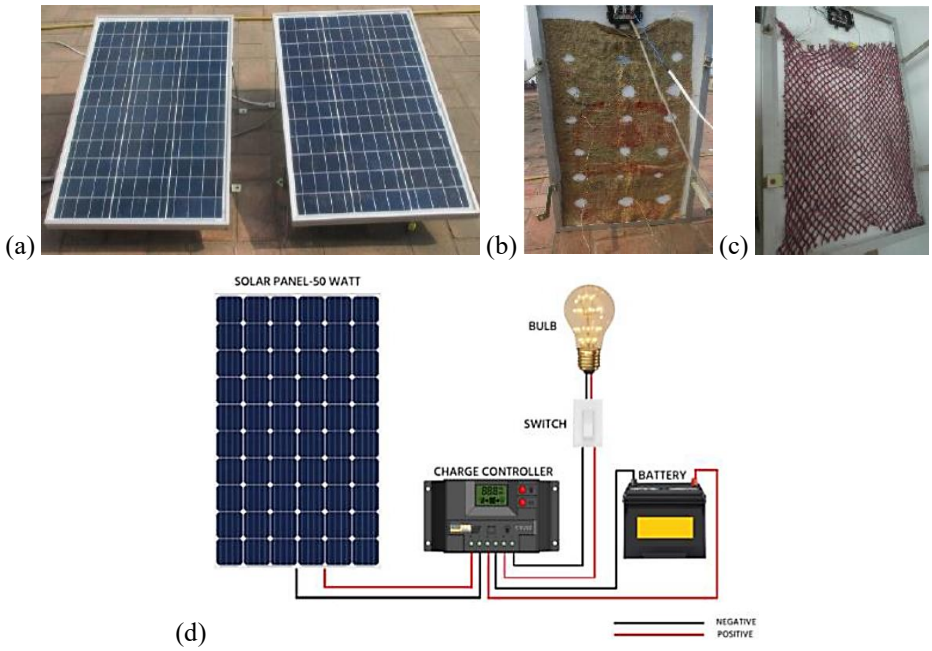


Figure 1. (a) Selected two PV modules setup (b) Back surface of PV module attached with jute wick and position of thermocouples (c) Back surface of PV module attached with wool wick (d) Schematic representation of the experimental setup of PV module

2.2. Test System

We carried out our tests on the rooftop of the institutional building of the Military Institute of Science & Technology, Mirpur Cantonment, Dhaka, Bangladesh which is 30 feet from the ground and the inclination angle of the panels was 23.45˚ to the south (Deb et al., n.d.). Most of the data were taken from March to August when the solar irradiance was high. K-type thermocouples were strategically positioned on the back surface of the PV modules with the help of glue guns to capture the working temperature. Jute and woolen fabrics were added to the back surface of the panels by glue. A digital thermometer was used to measure the temperature. A multimeter was used to measure both voltage and current. Two panels were kept parallel to each other and at different ambient temperatures, the module temperature, Current, and voltage of both panels were noted. Then one panel was normally water sprayed over the front surface. Similarly, wet jute fabric was added behind one panel and data were collected. It was aimed to obtain readings that directly correlate with the actual temperatures experienced by the PV modules during operation, rather than the ambient atmospheric temperatures which might not be representative of the module's working conditions. The process was repeated with wet woolen material too.

2.3. Data Collection

At different module temperatures (T) by measuring current(I) and voltage(V), Power Output (P1,P2,P3, P4) and Solar Efficiency (E1, E2,E3,E4) were obtained.

Table 2. Variation of Solar Cell Efficiency with Different Module Temperatures

T (Module) ℃	P1(Undisturbed) Watt	P2(Water Spray) Watt	P3(Wet Jute) Watt	P4(Wet Wool) Watt	E1(Undisturbed) %	E2(Water Spray) %	E3(Wet Jute) %	E4(Wet Wool) %
45	30.6	31.724	32.522	34.17	6.8	7.06	7.23	7.59
44	31.155	31.758	33.165	34.228	6.92	7.09	7.37	7.61
43	31.2	31.916	33.32	34.476	6.93	7.1	7.4	7.66
42	31.717	31.959	33.415	34.825	7.05	7.16	7.43	7.74
41	32	32.232	33.432	34.946	7.11	7.19	7.43	7.77
40	32.118	32.342	33.567	35.577	7.14	7.21	7.46	7.91
39	32.274	32.437	33.969	36.158	7.17	7.4	7.55	8.04
38	33.2	33.2831	34.272	36.256	7.38	7.53	7.62	8.06
37	33.734	33.864	34.34	36.72	7.5	7.53	7.63	8.16
36	33.936	33.9486	34.71	36.72	7.54	7.54	7.71	8.16
35	33.936	34.1112	35.223	37.128	7.54	7.58	7.83	8.25
34	34.672	35.292	35.828	37.43	7.7	7.84	7.96	8.32
33	35.119	35.8561	36.888	38.009	7.8	7.97	8.2	8.45
32	35.64	35.875	37.26	38.19	7.92	7.97	8.28	8.49
31	35.657	35.908	37.555	38.793	7.92	7.98	8.35	8.62
30	36	36.4416	37.769	38.964	8	8.1	8.39	8.66

2.4. Electrical Output (Theoretical)

Theoretical Power Output can be calculated with the help of the temperature coefficient. The Formula can be written as (EnergySage, n.d.)

$$P(theoretical) = \begin{cases} Maximum\ Power, Pmax - \\ (Module\ Temperature - Ideal\ Temperature) \times \\ Temperature\ Coefficient\ \times Pmax \end{cases} \quad (1)$$

Here, the Ideal Temperature means 25°C. The maximum power output and temperature coefficient of our Panel are 50 watt and 0.44%°C-1 respectively.

2.5. Solar Cell Efficiency

The formula can be written as (Kumar, 2017)

$$Solar\ efficiency = \frac{\left(Power\ Output\left(\frac{Theoretical}{Experimental}\right)\right)}{Solar\ Panel\ Area \times Solar\ Irradiance} \quad (2)$$

We know the value of Solar Irradiance varies from 950 to 1050 W/m2. To simplify the calculation we have used a Solar Irradiance value of 1000 W/m2.

3. Result

3.1. Power Output Increment

The graph illustrates the relationship between working temperature (35°C to 45°C) and power output under various conditions. The undisturbed baseline output remains almost steady at about 30 Watts. When subjected to water spray, the power output mirrors the baseline, indicating minimal impact.

However, the inclusion of wet woolen fabric behind the panel significantly boosts the output by approximately 1.5 to 3 watts across the temperature range, suggesting its potential to enhance module efficiency. The jute material's impact is less than woolen material achieving around 0.5 to 1.5-watt increment. The standout observation is the efficacy of wet woolen fabric in elevating power output. The findings in our study agree with the trend of power output increment in the previous studies by the application of wet wick cooling techniques on PV modules due to their higher moisture regain and moisture content properties (Bhakre et al., 2023, Alktranee & Bencs, 2023).

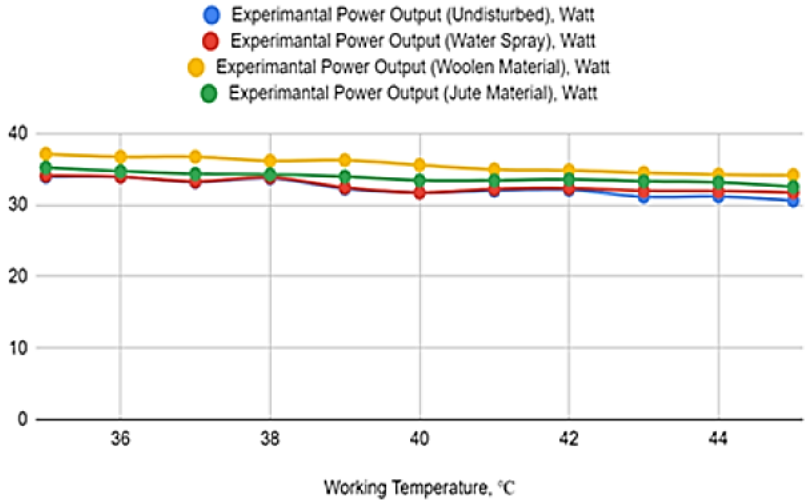


Figure 2. Variation of power output with different temperatures

3.2. Solar Efficiency Increment

The depicted graph portrays the relationship between working temperature, ranging from 35°C to 45°C, and solar cell efficiency under various conditions. The theoretical efficiency, represented by the blue line, serves as a benchmark, consistently hovering around 10%. The undisturbed panel's experimental efficiency, shown by the red line, remains relatively stable just below 8%. Introducing water spray sees efficiency, indicated by the yellow line, aligns closely with the undisturbed panel, suggesting that water spray alone has minimal effect on efficiency. Remarkably, the presence of wet woolen fabric behind the panel enhances the solar cell's efficiency, as evidenced by the green line, which consistently outperforms the undisturbed condition. In comparison, the jute material, denoted by the orange line, exhibits efficiency values less than the woolen material. The primary takeaway is the pronounced positive effect of the wet woolen and jute fabric on solar cell efficiency, warranting further exploration into its potential benefits for solar applications. Previous studies (Said et al., 2023,

Bhakre et al., 2023, Alktranee & Bencs, 2023) also support the trend of efficiency of increment in our study by introducing wet jute and wool wicks cooling techniques.

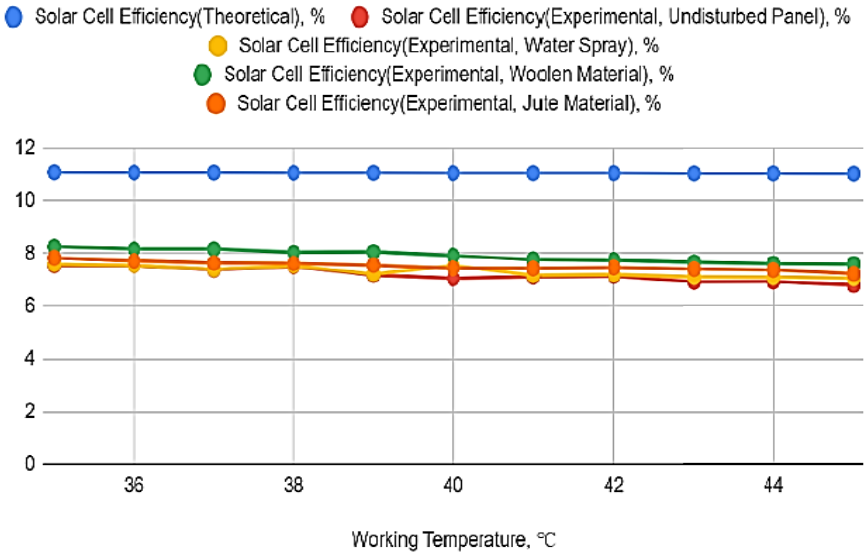


Figure 3. Variation of Solar Cell efficiency with different temperatures

4. Conclusion

PV Modules demonstrate optimal efficiency at 25°C. As the temperature rises, especially beyond 40°C, there's a significant decline in solar power output. However, when these high-temperature modules were subjected to cooling via a frontal water spray, an increase in power output of approximately 0.4 watts was observed in this study. Incorporating wet jute and wet woolen fabrics as cooling mediums produces even more yielded enhancements, with an increase of around 1.5 watts and 3.0 watts respectively. In terms of solar efficiency, wet woolen fabric takes the lead with an enhancement of about 0.7%. This is followed by jute, which offers an improvement of approximately 0.35%, and the water spray method at a modest 0.1%. Both woolen and jute materials present themselves as effective strategies to amplify solar panel efficiency, with the selection contingent on specific objectives and financial considerations. Future studies might consider year-round testing to discern seasonal variations in power output. Moreover, there's potential in exploring a combined cooling approach that employs both wet woolen and jute materials at the back of the module, paired with a frontal water spray. Additionally, examining the performance of larger power output PV modules in subsequent studies would provide a broader perspective on scalability and efficiency gains.

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